ALEX MERCER (USER #2 DEMO)

SENIOR 3D CAD/CAM DESIGNER & CNC MANUFACTURING LEAD

user2@example.com • 021 000 0000 • Auckland, New Zealand

PROFESSIONAL SUMMARY

Hands-on 3D CAD/CAM Specialist and CNC Workshop Manager with deep expertise bridging complex 3D surface modelling and precision digital fabrication. Proven track record of translating architectural artist concepts into production-ready 3D building facade geometry, engineering high-performance yachting components for America's Cup campaigns, and leading end-to-end marine CNC machining operations. Experienced in managing distributed international design workflows, including leading remote CAD/CAM direct reports in Europe while running local workshop production in New Zealand.

WORK EXPERIENCE

Marine Engineering & CNC Fabrication Workshop **2023 – Present**

CNC Workshop Manager & Lead 3D CAD/CAM Technician Auckland

- **End-to-End 3D Design to Fabrication:** Lead both 3D CAD modelling and multi-axis CAM toolpath programming through to hands-on CNC machine operation for precision marine engineering and bespoke structural components.
- **International Remote Team Leadership:** Manage and review daily CAD/CAM deliverables for a direct report working remotely from Poland, coordinating asynchronous 3D file handoffs, quality control, and production scheduling.
- **Workshop Operations & Optimization:** Oversee machine scheduling, tooling selection, material nesting, and quality assurance to hit tight marine refit and custom manufacturing deadlines.
- **Design for Manufacturability (DFM):** Partner directly with naval architects, engineers, and project managers to refine 3D geometries for optimal CNC machining efficiency and structural tolerance.

Architectural Facade & Bespoke 3D Modelling Studio **2020 – 2023**

Senior 3D CAD & Facade Modeller Auckland

- **Concept-to-CAD Facade Engineering:** Translated 2D artist impressions and architectural concept sketches into millimeter-accurate 3D models for complex commercial building facades and bespoke architectural elements.
- **Fabrication-Ready Geometry:** Developed parametric and complex surface models engineered specifically for downstream CNC milling, mould-making, and structural assembly.
- **Multi-Disciplinary Coordination:** Collaborated with architects, structural engineers, and fabrication teams to resolve geometric clashes and deliver complete 3D production packages.

Emirates Team New Zealand (America's Cup Campaign Contractors) **2018 – 2020**

3D CAD Modeller & High-Performance Marine Component Specialist Auckland

- **America's Cup 3D Part Modelling:** Created precision 3D CAD models for custom high-performance yachting parts, aerodynamic/hydrodynamic tooling, and composite mould assemblies for New Zealand's America's Cup team.
- **Ultra-Tight Tolerance Execution:** Delivered complex 3D surface and solid models under high-pressure campaign timelines where weight, strength, and sub-millimeter CNC accuracy were mission-critical.

TECHNICAL SKILLS & CERTIFICATIONS

Technology & Tools: 3D CAD Surface & Solid Modelling (Rhino 3D, Grasshopper, SolidWorks, Fusion 360, AutoCAD, Inventor), CAM Toolpath Programming (Mastercam, PowerMill, Fusion CAM), Multi-Axis CNC Routing & Milling, G-Code, DFM, PDM/Cloud CAD Collaboration.

Core Competencies: Concept-to-3D Reconstruction, Architectural Facade Modelling, Marine & Composite Tooling, CNC Workshop Management, Remote Team Leadership (NZ-Europe Async), Precision Quality Control

Certifications & Highlights: Trade & Advanced CAD/CAM Fabrication Experience, America's Cup High-Performance Marine Manufacturing

Interests: Yachting & Marine Design, Digital Fabrication, 3D Printing, Industrial Design

EDUCATION

New Zealand Engineering & Design Training | 3D CAD/CAM Design & Precision CNC Manufacturing Auckland